

NTAREI Phase 1 Meeting Check-In

Functional / Technical Specification

Offline local tablet application • Phase 1 only

1. Objective and Scope

Replace the current paper-and-pen guest sign-in sheet with a very small tablet application that collects attendee information locally, displays the completed information for registration staff, and exports the collected records after the meeting.

IN SCOPE	OUT OF SCOPE
• Offline operation; no Internet required during meeting	• Advance registration
• Local storage on each tablet	• QR codes or tickets
• Attendee sign-in form	• Online payments or Square integration
• First-time attendee indicator	• Tablet-to-tablet synchronization
• Marketing-source question	• Cloud/database server
• Read-only staff handoff screen	• Membership lookup
• Edit/back function before finalizing record	• Phone-number collection
• Next Attendee function	• Email-address validation beyond basic required-field handling
• CSV export of local records	• Duplicate detection or merging
• iPad Mini testing and Android-tablet compatibility	• User accounts/login
	• Analytics/dashboard beyond exported data

2. Functional Requirements

2.1 Attendee Sign-In Screen

The application opens directly to a simple, touch-friendly sign-in screen. The attendee must not see prior attendee records.

Welcome
NTAREI MAIN MEETING
Please sign in.

First name [_____]
Last name [_____]
Email [_____]

Is this your first time to attend an NTAREI Main Meeting?
☐ Yes, my first time.

How did you learn about this meeting?
☐ Google search ☐ ChatGPT / AI search
☐ Saw video ad ☐ Someone told me
☐ NTAREI email ☐ Facebook

[] Meetup
[] Other: [_____]

Thank you. We're glad you came.

[CONTINUE →]

- First Name, Last Name, and Email are required for a normal submission.
- The first-time checkbox is optional; unchecked means no first-time indication is recorded.
- Marketing source should be single-select for Phase 1 unless implementation is materially simpler as multi-select. Preferred behavior: one source plus optional Other text.
- If Other is selected, the attendee may enter free text.
- No phone-number field.
- No attempt is made to correct misspellings or validate whether an email address is genuinely deliverable.
- The layout must fit comfortably on an 8-inch-class tablet in portrait orientation and remain usable on the user's iPad Mini.

2.2 Continue / Staff Handoff

When the attendee taps CONTINUE, the app switches to a read-only handoff screen. The entered information remains clearly visible so registration staff can make the attendee's badge and complete any manual desk procedures.

Suggested handoff screen:

MARK WHITTED

mark@example.com

First NTAREI Main Meeting: YES

How learned: ChatGPT / AI search

← EDIT

RETURN TO STAFF →

The words "RETURN TO STAFF →" are instructional text for the attendee. They do not advance or erase the record.

2.3 Staff Actions

- EDIT / BACK returns to the attendee form with the current values preserved.
- NEXT ATTENDEE is a staff action shown on the staff/handoff screen in a clear but non-ambiguous location.
- When staff taps NEXT ATTENDEE, the current record is saved (if not already saved), the form is cleared, and the attendee sign-in screen is displayed for the next person.
- The application must not require a hidden, faint, or secret control for normal staff operation.

2.4 Data Export

A staff-accessible Export function must create a CSV file containing all records stored on that tablet. Export is performed after the meeting or as needed for backup. Each tablet exports its own file; Phase 1 does not synchronize tablets.

Field	Purpose
record_id	Locally generated unique identifier
created_at	Local date/time record was completed
first_name	Attendee first name

last_name	Attendee last name
email	Attendee email
first_time	Boolean / Yes-No
source	Google / ChatGPT-AI / Video ad / Someone told me / NTAREI email / Facebook / Meetup / Other
source_other	Free text when Other is used

3. User Workflow

1. Staff opens the NTAREI Check-In app on each tablet before registration begins.
2. Staff hands a tablet to an arriving attendee, similar to the current clipboard process.
3. Attendee enters name, email, first-time status, and how they learned about the meeting.
4. Attendee taps CONTINUE.
5. App shows the attendee information read-only and instructs the attendee to RETURN TO STAFF.
6. Attendee returns the tablet while staff uses the displayed information to prepare the badge and perform any other manual registration tasks.
7. If a correction is needed, staff taps EDIT / BACK, corrects the information, and returns to the handoff screen.
8. When finished with that attendee, staff taps NEXT ATTENDEE.
9. App saves the record and returns to a blank attendee form.
10. At the end of the meeting, staff exports a CSV from each tablet and combines/imports the files as desired.

4. Technical Specification

4.1 Architecture

- Single-page web application (SPA) / Progressive Web App (PWA).
- HTML, CSS, and JavaScript/TypeScript; no server-side component required for Phase 1.
- Installable to the Home Screen and launchable in standalone/app-like mode where supported.
- After initial installation/caching, the app must operate with Wi-Fi and cellular data disabled.
- Each tablet is independent. All meeting data remains local until explicitly exported.
- Primary local persistence: IndexedDB. Do not use cookies as the data store. localStorage may be used only for trivial app preferences if needed.
- PWA service worker caches the application shell/assets for offline launch.
- No external JavaScript/CSS/CDN dependencies at runtime. Any dependency must be bundled locally so offline operation is deterministic.

4.2 Data Safety

- Do not delete stored records when the form is cleared.
- NEXT ATTENDEE must clear only the working form state after the current record has been persisted.
- CSV export must quote/escape commas, quotes, and line breaks correctly.
- Record IDs must be unique enough to permit later merging of exports from multiple tablets.

- No attendee should be able to browse prior attendees from the attendee-facing form.
- Provide a staff-only way to view record count and export data. Phase 1 may use a simple obvious staff control rather than authentication; do not invent complex security.

4.3 UI / Device Requirements

- Portrait-first responsive design; usable from approximately 7–8 inches through standard iPad sizes.
- Large touch targets (approximately 44 CSS px minimum where practical).
- Large readable labels and text inputs suitable for a standing attendee.
- Native on-screen keyboard should appear naturally for text/email inputs.
- No horizontal scrolling.
- No requirement for keyboard, mouse, stylus, or external hardware.
- Test first on the user's iPad Mini; verify on the selected Android tablet before meeting use.

4.4 Suggested Project Structure

```
ntarei-checkin/
  index.html
  src/
    app.js (or app.ts)
    db.js
    export.js
    styles.css
  manifest.webmanifest
  service-worker.js
  icons/
  README.md
  tests/
```

5. Acceptance Criteria

- App installs/launches successfully on the test iPad Mini.
- App launches and remains fully functional with Wi-Fi turned off after installation/caching.
- Attendee can complete the defined form without seeing any prior attendee data.
- CONTINUE shows the current attendee's information read-only.
- EDIT returns to the populated form without data loss.
- NEXT ATTENDEE persists the record, clears the form, and returns to the sign-in screen.
- At least 30 consecutive test registrations can be entered without failure or noticeable degradation.
- Closing and reopening the app does not erase saved registrations.
- CSV export contains all saved test registrations and opens correctly in Excel/Numbers.
- Two tablets may be used simultaneously with no network connection; each maintains its own independent records.
- No Internet/network request is required during normal meeting operation.

6. Phase 1 Test Plan

1. Install app on iPad Mini; open it once while online if required for PWA installation/cache.
2. Enable Airplane Mode and explicitly turn Wi-Fi off.
3. Launch app from Home Screen.
4. Enter 30 fake attendees at realistic typing speed, including long names and email addresses.

5. Exercise each marketing-source choice and Other text.
6. Use EDIT on several entries and verify changes are saved.
7. Close the app completely after several records, reopen, and verify records remain.
8. Export CSV and verify row count and field values.
9. Repeat critical flow on the purchased Android tablet.
10. Before live use, clear all test data and verify zero local records.

7. Claude Code Prompts

Use the prompts below sequentially. Keep Claude Code tightly constrained to Phase 1. Do not allow it to add servers, authentication, QR features, payments, cloud sync, frameworks, or unrelated abstractions unless explicitly approved.

Prompt 1 — Build the Phase 1 app

You are implementing a very small offline-first tablet check-in application for NTAREI. Read this entire specification before changing anything.

GOAL

Build Phase 1 only: an installable offline PWA that collects attendee sign-in information locally on one tablet, shows a read-only staff handoff screen, then resets for the next attendee. Multiple tablets operate independently. There is no server and no live synchronization.

FUNCTIONAL FLOW

1. Attendee screen:
 - Welcome / NTAREI MAIN MEETING / Please sign in.
 - First name (required)
 - Last name (required)
 - Email (required; use email input type, but do not build elaborate validation)
 - Checkbox: “Yes, my first time.” for “Is this your first time to attend a NTAREI Main Meeting?”
 - “How did you learn about this meeting?” with one selection from:
 - Google search
 - ChatGPT / AI search
 - Saw video ad
 - Someone told me
 - NTAREI email
 - Facebook
 - Meetup
 - Other
 - If Other is selected, show/enable a free-text field.
 - Thank-you text: “Thank you. We’re glad you came.”
 - CONTINUE button.
2. CONTINUE opens a read-only handoff screen showing all entered information clearly. It must include instructional text “RETURN TO STAFF →”.
3. Staff actions on the handoff screen:
 - EDIT / BACK: return to the populated attendee form.
 - NEXT ATTENDEE: persist the finalized record, clear working state, and return to a blank form.
 - Do not implement a hidden/faint/secret normal-use control.
4. Local data:
 - Persist finalized records in IndexedDB.
 - Fields: record_id, created_at, first_name, last_name, email, first_time, source, source_other.
 - Data must survive app close/reopen.
 - Do not expose prior attendees to the next attendee.
5. Staff utility:

- Provide a simple staff-accessible record count and Export CSV function.
- CSV must be standards-compliant and safely escape commas, quotes, and line breaks.
- No login/authentication system for Phase 1. Keep the utility unobtrusive but maintainable.

TECHNICAL CONSTRAINTS

- Offline-first PWA; after initial install/cache, all normal operation must work with Wi-Fi disabled.
 - No backend/server/database service.
 - No runtime CDN calls or externally hosted dependencies.
 - Bundle everything needed locally.
 - Prefer plain HTML/CSS/JavaScript (or very lightweight TypeScript only if already configured).
- Do not introduce React/Vue/etc. merely for this app.
- Service worker caches app shell/assets.
 - manifest.webmanifest supports Home Screen installation.
 - Portrait-first responsive UI for 7–11 inch tablets, especially iPad Mini.
 - Touch targets roughly 44 CSS px minimum. No horizontal scrolling.
 - No phone number. No preregistration. No QR. No payment/Square integration. No cloud sync. No duplicate detection. No membership lookup.

DELIVERABLES

- Working app
- README with exact local-development, install, offline-test, data-clear, and CSV-export instructions
- Automated tests where practical for persistence/export/business logic
- A short manual test checklist covering iPad/iOS and Android/Chrome

Before coding, inspect the repository and summarize the minimum implementation plan. Then implement it. Keep the solution intentionally small.

Prompt 2 — Review for scope and offline reliability

Review the completed NTAREI Phase 1 check-in app against the specification. Do not add new product features.

Focus on:

1. Does every required attendee field and marketing-source choice exist?
2. Is the attendee → CONTINUE → read-only handoff → RETURN TO STAFF → staff EDIT/NEXT ATTENDEE flow correct?
3. Is the record guaranteed to be persisted before working state is cleared?
4. Can prior attendee information ever be exposed to the next attendee? Fix that if possible.
5. Does IndexedDB data survive close/reopen?
6. Does the service worker allow the app to launch and function with Wi-Fi fully disabled after the first successful install/cache?
7. Are there any CDN, analytics, font, API, or other network dependencies that would break strict offline operation? Remove them.
8. Does CSV export properly escape fields and include exactly the specified columns?
9. Is the UI practical on an iPad Mini in portrait orientation with large touch targets and no horizontal scrolling?
10. Is any implementation unnecessarily complex for Phase 1? Simplify where safe.

Run the available tests. Add only tests that directly protect Phase 1 behavior. Report defects found, changes made, and any remaining device-specific risks.

Prompt 3 — Prepare the iPad Mini acceptance build

Prepare the NTAREI Phase 1 check-in app for acceptance testing on an iPad Mini. Do not expand scope.

Tasks:

- Verify PWA manifest, icons, standalone display behavior, viewport settings, and service-worker registration are appropriate for current iPadOS Safari.
- Ensure the app can be loaded/installed once and then used with Wi-Fi disabled.
- Ensure email fields invoke the appropriate mobile email keyboard where the browser supports it.
- Ensure form controls are comfortably usable in portrait orientation and are not obscured by the on-screen keyboard.
- Ensure reopening the app cannot silently destroy IndexedDB records.

- Add a clearly documented SAFE procedure to export test data and then clear all local test data before live use. Clearing data must require an explicit staff action and confirmation.
- Update README with concise step-by-step iPad installation and offline acceptance-test instructions.
- Run all tests and provide the exact manual acceptance sequence for 30 fake attendee entries, app close/reopen, CSV export, and final test-data clearing.

Do not add a backend, authentication, QR, payments, synchronization, preregistration, or any other Phase 2 functionality.

Prompt 4 — Android tablet compatibility check

Without changing the Phase 1 requirements, review the NTAREI check-in PWA for compatibility with a low-cost current Android tablet using Chrome.

Check:

- Install/Add to Home Screen behavior
- Offline launch after initial caching
- IndexedDB persistence
- Touch target sizes and portrait layout on roughly 7-10 inch screens
- On-screen keyboard behavior for name and email fields
- CSV download/export behavior
- App close/reopen persistence
- No runtime network dependencies

Make only compatibility fixes that preserve the existing iPad behavior and keep the codebase simple. Do not add new product features. Update the README with Android installation/offline testing instructions and list any limitations that must be tested on the actual purchased tablet.

8. Deferred Phase 2+ Ideas (Not for Current Build)

The Phase 1 architecture should not deliberately prevent later expansion, but no code should be added now solely for these future ideas: preregistration on NTAREI.com, shared person/event/attendance database, QR check-in, attendee lookup, payment status/Square integration, membership status, cross-tablet synchronization, reporting, or marketing automation.

9. Phase 1 Decision Summary

- Tablets function as electronic clipboards, not fixed registration kiosks.
- Two tablets are sufficient for initial use; attendees can fill them out while staff processes another person.
- App is locally functional and does not depend on venue Wi-Fi.
- Each tablet stores its own data independently and exports its own CSV.
- The attendee sees a normal CONTINUE button; no secret/faint control is required.
- The completed attendee information remains visible for staff badge preparation before staff starts the next attendee.
- KISS applies to the attendee workflow and the implementation—not merely to minimizing lines of code.